[Total No. of Question -04] [Total No. of Pages - 02]

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T.Y B. Tech (AIDS/AIML/AI/DS)

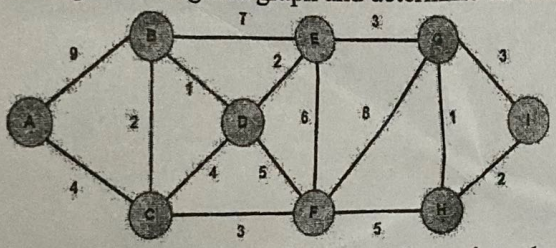
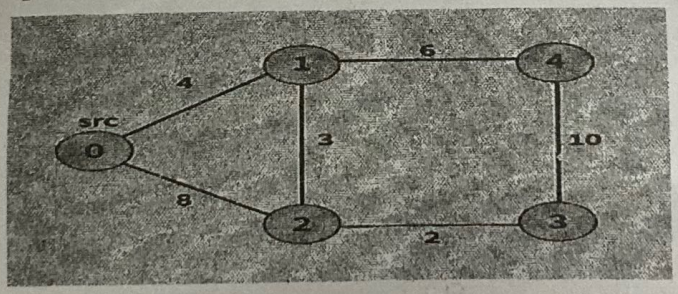
CAE II- WINTER 2026

Design and Analysis Algorithm (25BTAIPCL502)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

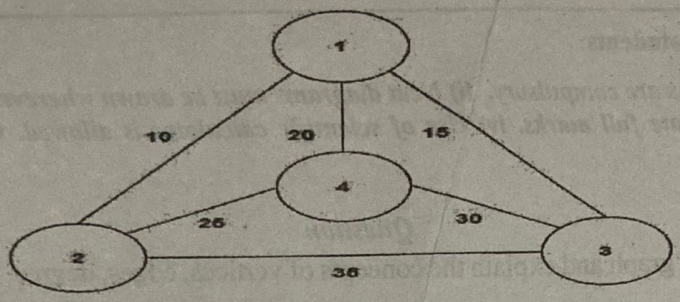
i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed. v) Assume suitable data, if necessary.

Q. No		Marks	CO	BL
	Question			
Q 1. (a)	Define a graph and explain the concepts of vertices, edges, degree, paths, cycles, connected graphs, and weighted graphs.	[5]	3	L1
(b)	Explain Prim's algorithm for finding a Minimum Spanning Tree of a connected weighted graph.	[5]	2	L2
	OR			
Q 2. (a)	Apply Kruskal's algorithm to construct the Minimum Spanning Tree for a given weighted graph and determine its total cost.  Apply Dijkstra's algorithm to determine the shortest path from a specified source vertex to all other vertices in a given weighted graph.	[5]	3	L3
(b)		[5]	3	L3
Q 3. (a)	Explain the Matrix Chain Multiplication problem and why the order of matrix multiplication affects the number of scalar multiplications.	[5]	2	L2
(b)	Explain the concept of an Optimal Binary Search Tree and describe why it is useful when search frequencies are known.	[5]	2	L2
	OR			

[Total No. of Question -04] **[Total No. of Pages - 02]**

Apply the Dynamic Programming approach to determine the minimum-cost tour for a given Travelling Salesman Problem.

Q 4. (a)



[5] 2 L3

(b) Explain the difference between Dynamic Programming and Divide-and-Conquer techniques with suitable examples. [5] 2 L2